Yancheng (Jason) Liu

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Education

Imperial College London *Sep 2026–Sep 2027*
MRes in AI and Machine Learning
- Selected courses (2026–27): Machine Learning Systems and Hardware; Natural Language Processing.

University of Cambridge, Selwyn College *Oct 2023–Jun 2026*
BA (Hons), Natural Sciences; Part II specialisation in Astrophysics
Upper Second-Class Honours.
- Class by year: Year 2 — First Class; Years 1 and 3 — Upper Second (II.1).
- Selected courses: Part IA Mathematics, 69.2/100; Part IB Mathematics, 75.2/100 (First Class).
- Part II computational projects: 120/120.

Publications

Yuhan Zhu*, **Yancheng Liu***, Chuheng Du, Junyi Chen, Shuai Huang, Tailin Zhou, Shengzhong Liu, Fan Wu, and Guihai Chen. Racer: Recoverable KV Cache Management for Multi-Turn LLM Agents. *Under review at* **ICLR 2027**. *Equal contribution.

Research Experience

Racer: Recoverable KV-Cache Management for Multi-Turn Agents *Aug–Sep 2026*
School of Computer Science, Shanghai Jiao Tong University
Supervisor: Prof. Shengzhong Liu / ***Co-first author*** / *Under review at ICLR 2027*

- A wrong action taken under KV compression changes the environment, and the error accumulates over later turns. We proposed Racer, which detects risky drafts before an action executes, restores the original evidence and regenerates; it can be layered on top of existing KV-compression methods.
- Across six KV-compression backends under a fixed history budget, Racer raised mean task success on BFCL Long Context by 43.7% (relative); under concurrent serving, it reached 1.67× the task throughput of full-context execution.
- Split the history into offloaded original text, compressed KV and a few events kept verbatim, all under one budget; a linear risk detector reads the prefill hidden state and draft signals to decide whether to restore evidence. Trained the gist embeddings and QKV projections of the group's compressor, C^2KV (arXiv:2607.17715), on correct tool-call continuations. Implemented the concurrent serving runtime and session-owned compressed KV in SGLang, and the eager, SDPA, FlexAttention and FlashAttention-2 mask backends.

Mechanistic Interventions and Failure Signals in Multimodal Models *Jun–Sep 2026*
Machine Intelligence Laboratory, University of Cambridge
Supervisors: Dr Mengjie Qian and Prof. Kate Knill / *Manuscript in preparation for Interspeech 2027*

- MC dropout ranks wrong predictions first, and a gated cross-attention model trained on word–audio aligned input improves, but it is unclear whether these come from particular layers, the input or training. We intervened on the internal representations and inputs of both models to attribute these signals and gains to their sources.
- With the classifier's predictions and errors fixed, MC-dropout mutual information raised error-detection AUROC by 0.07–0.12 under all three training objectives. We proved that it measures the margin fluctuation induced by dropout, set layer-adaptive dropout rates accordingly, and further split the effect by layer: perturbing only the lower half of the blocks matched all layers, while perturbing only the upper half gave almost no gain.
- Cross-replay (scoring each model of a pair on both inputs) showed that the gain apparently due to alignment came from training and checkpoint selection. This is because the global gated cross-attention block is invariant to permuting its audio rows: shuffling the word–audio correspondence changes its output only within floating-point error.

ShapeFlow: Serving Agentic Workflows with Late-Bound Representations *May–Jul 2026*
School of Computer Science, Shanghai Jiao Tong University
Supervisor: Prof. Shengzhong Liu
shapeflow-p1-study

- Before passing retrieved web pages on, deep-research agents have the LLM decode a summary of each page, and this serial decoding is the main inference bottleneck. I designed ShapeFlow, a scheduler that decides dynamically under engine load between per-page summaries and a decision model that outputs only span IDs, and makes this choice greedily together with request admission, speeding up inference.
- On BrowseComp-Plus, CPU-selected spans in place of per-page summaries cut vLLM busy time per task by 67%; running outside an unmodified vLLM, the scheduler drove live agents end to end.
- Added hooks to open_deep_research that switch the summarization step to the decision model, and tagged and timed every vLLM request; the scheduler admits plans in increasing order of estimated token cost, at most one per job.

Product-Space Orthogonalization for Transformer Optimizers *Mar–May 2026*
School of Artificial Intelligence, Shanghai Jiao Tong University
Supervisors: Yiting Chen and Prof. Junchi Yan
coupled-muon-nanogpt

- Muon orthogonalizes each matrix's update separately, but attention logits depend on the Q–K product (outputs on the V–O product), whose update is left uncontrolled. I designed Coupled Muon to orthogonalize the product's update.
- Across 960 runs on a 60M-parameter ladder, it lowered validation loss by 0.007–0.024 nats relative to Muon on every RoPE dense and MoE variant, but not on MLA; the gain sat in the per-head Q–K and V–O pairs, with a lower maximum attention logit.
- Preconditioned each factor with its partner's current value inside Newton–Schulz, a Stiefel-limit approximation of steepest descent under a product spectral-norm constraint. Also derived TangentMuon, which takes the matrix sign of the product's first-order change and splits it back to the two factors through a damped Sylvester equation.

Industry Experience

Research Intern — Speech & Audio Research *Dec 2024–Apr 2025*
Lenovo Audio Research Lab, Shenzhen
Supervisor: Zhuochao Zheng

- Developed a GTCRN-based experimental framework for lightweight causal speech enhancement in streaming settings, with waveform-reconstruction training and STOI-based evaluation for speech intelligibility and checkpoint selection.

Software Engineering Intern *Aug–Oct 2023*
Lenovo, Vantage AIGC Team

- Co-developed a Python/LangChain assistant connecting internal knowledge retrieval, APIs, and database queries through RAG and agent tools.

Awards

- Selwyn College Distinguished Student Award; Tim Langley Bursary (£2,000), 2025.
- BPhO Top Gold, 2023; UKChO First Place in China, 2022.
- S.T. Yau High School Science Award, Regional Second Prize, 2021.